

Roll No.: 1068899

Subject : 88702

PART - A

University of Mumbai

(Read the Instructions given on the reverse side)
Use Blue/Black Ball Pen only to darken the appropriate circles. Do not fold, tear, wrinkle, staple or use whitener on the cover page of the answer book. Do not write or mark on the barcode and the timing tracks.

BCNO 259617344

Do not write/Mark on Bar Code

Supplement Booklet Information

Supplement Taken: ☒ Yes ☐ No

If Yes, How Many Booklets: ☐ 1 ☐ 2 ☐ 3 ☐ 4

Write the Supplement Barcode No.:

Signature of Candidate: *Tanish* Date: 22/04/2024

Exam Date: 22 04 2024

Program Name: TY BscIT Sem VI

Program Code: 1S00256

Year: 2024

Semester: 0006

Subject Name: Security in Computing

Subject Code: 88702

PRN: 2021016401651294

Candidate Seat No.: 1068899

College Seal

Q.P. Code (Printed on Question Paper): 52552

College Code: 0229

Signature & Name of the Block Supervisor with Date: *Bhushan Chaudhary* 22/04/24

Signature of Chief Coordinator: *Fernandes*

12379798

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1068899

22-04-2024:02:30:PM-05:00:PM

1. 1S00256:T.Y. B.Sc. (I.T.)

2. SEMESTER Sem VI

3. 88702::Security in Computing

0229:ST.ANDREW'S COLLEGE OF ARTS, SCIENCE & COMMERCE BANDRA (W).

0229/22-04-2024/15

Student's Signature: *Tanish*

The data entered by the Candidate should be verified by the Block Supervisor.

Part - B

BCNO 259617344

Do not write/Mark on Bar Code

Question wise marks given by Examiner

Q. No.	Marks	8	9
1			
2			
3			
4			
5			
6			
7			
Total			

Name and Signature of Examiner with Date

Exam Date: 22/04/2024

Program Code & Name: 1S00256 TY BscIT

Subject Code & Name: 88702 / SIC

Year / Semester: 2024 / 06

If Physically challenged: LD ☐ PC ☐ VI ☐ Regular ☐

Question wise marks given by Moderator

Q. No.	Marks	8	9
1			
2			
3			
4			
5			
6			
7			
Total			

Name and Signature of Moderator with Date

INSTRUCTIONS TO CANDIDATES

1. Candidates should occupy the correct seats and write correct seat number and other details in the space provided for the purpose on the answer-books.
2. Candidates who are not in their seats by the time notified, will not as a rule be permitted to appear for the examination. The Senior Supervisor may at his/her discretion admit those who give him/her a satisfactory reason.
3. Each answer-book contains Thirty two pages. Check whether the pages are properly numbered.
4. Candidates should write their answers in legible handwriting. They are warned that zero marks may be assigned to answers which cannot be assessed by the examiners owing to illegible handwriting.
5. Write on both sides of a page. Rough work where necessary, should be done on the last page in the space provided. No page should be left blank. Any such act shall be treated as unfair means.
6. Do not write anything in the Examiner & Moderator sheet (Part-B) & Re-Evaluator Sheet except Candidate details.
7. Do not damage or make any stray marks on the barcodes.
8. Candidates will not be permitted to leave the examination hall until half an hour after the question paper is distributed.
9. All answer-books supplied shall be returned whether written or blank. Nothing shall be written on the question paper.
10. No sheet shall be torn from the provided answer-books nor shall additional papers attached to them.
11. Even if it is mentioned in question paper to write each section in separate answer book, if any paper / subject have multiple sections, the candidate has to write all sections in one and the same answer book.
12. Write correct Barcode Supplement on main Answer Book.
13. A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer books to the Junior Supervisor. They should not leave their seats until answer-books from all candidates are collected by the Junior Supervisor.

UNFAIR MEANS IN THE EXAMINATIONS

14. Candidates shall write the answers only with BLUE/BLACK Ink Ball pen only. Use of any other Pen like Gel Ink or Fountain Ink or any other colour ink, will be treated as unfair means in terms of revealing of identity.
15. Candidates are forbidden to (i) bring any book, notes, scribbling papers, pages, Mobile phones/smart watches or any other similar devices. (ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
16. A candidate who disobeys any instructions issued by the Senior/ Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him / her by the University as per O-5050
17. Do not fold the answer book anywhere because it will be treated as unfair means in terms of revealing of identity.
18. Candidates suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the University in respect of the reported act of unfair means is binding on them/Exchange of writing materials, stencils, mathematical instruments etc. is strictly prohibited. If candidates want anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seats on any account.
19. Any method to bribe the examiner/s by attaching currency notes or letters is strictly prohibited and will result in serious action being taken by the University
20. Seat number should be written only the space provided for the same. Candidate should not write his/her name in any part of the answer-book. Writing Name, Seat No., Phone/Cell No., putting signature, use of religious invocation or any writing that is not relevant to the answers anywhere in the answer-book will be treated as attempts to reveal identity.
21. Underlining of answers for focusing attention is permitted. However, use of varied inks, except for illustrations and figures must be avoided. DO NOT use symbol like encircling the question or using colour arrows for P.T.O. These will all be considered as attempt to readily identify the specific answer-book & will be treated as unfair means.
22. Candidates are requested to go through the provisions of O-5050 for preventing malpractices at University.

IT IS PRESUMED THAT CANDIDATE HAS READ ALL THE ABOVE INSTRUCTIONS.



Space for Marks	Question No.	
	Q1.a	Three Ds of security in the context of your home and organization • In the context of 3Ds of security there are use 3Ds for give a well define security • In an organization we can secure with help of 3Ds of security → ① Detect ↓ ② Defance ③ Detrence • Detect:- in the context of Detect first the virus or any harmful thing entering in the organization ✓ it will be detected by some guard which is present at the gate. they will not allow to enter the organization without permission.

Space
for
MarksQuestion
No.

② Defiance:- It is second D of 3Ds. in case of entig the harmful thing in the organization first some Banior will Protet them Example. firewall.

③ Detrence:- It is the last and third 3rd D of 3Ds security.

✓ If the harmful thing ^{has} been entered in the organizat How can we remove and protect from this

• Detrence search the better way to recover itself. In every time and start improving all thing which is Damages. in the organizat

Q.1A - 2

Space
for
MarksQuestion
No.I
C

Defence Model:- Defence model is use to protect the sensitive data from outsider like hacker and viruses

- There are two Defence Model in Computing security

① Calipor Defence

✓ ② Onion model.

- we will understand onion model by the using some layer diagram





Space for Marks	Question No.
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- Firewall is a just like our house gate. It protect from outsider attacker. It check all the activity which are doing any thing will be enter in the host that time he check
- IPS: it is a second layer of Onion Model. If any attacker break the firewall it come into this security phase. and it protect that
- Authorization: it is third phase of onion model. in this phase we check the come person are authorized or not
- Authentication: it is a 4th phase of onion model that is use to Authenticate the person,
- Asset: is sensitive data which is present in last layer of the onion

Q.1C - 2

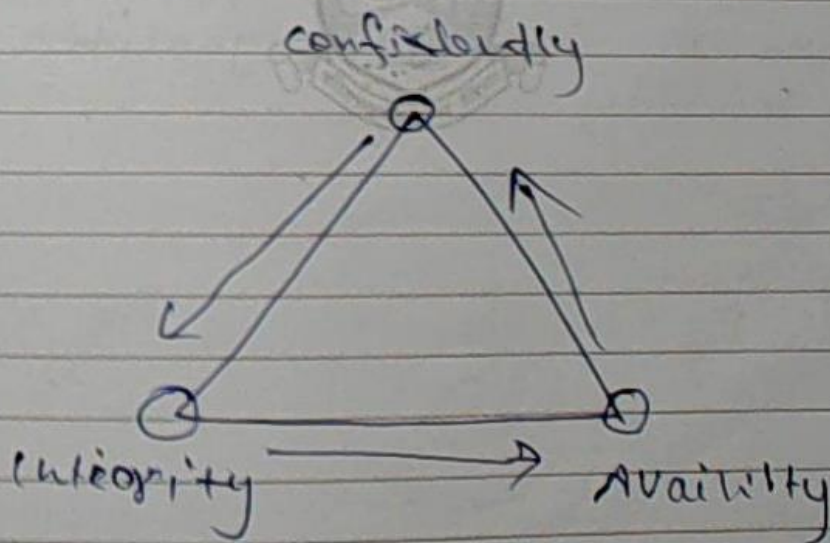
Space
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MarksQuestion
No.

1.8. CIA stands for Confidentiality, Integrity, Availability

→ CIA is use for securing the sensitive data and protecting from the hacker, and users

✓ CIA:- It is a best method for protecting data in computer related things

→ CIA:- can be defined by the using this diagram



→ Confidentiality:- In the Confidentiality the sensitive data like password and ATM pin will be store because it is very sensitive data

Space
for
MarksQuestion
No

→ Availability: Some Authorized person can be access the data no one can touch any inter net information.

→ Integrity: in the integrity all the information is present but valid people can access and update the information

Q.1F - 2

→ CIA is very powerful in security that provide best quality of security



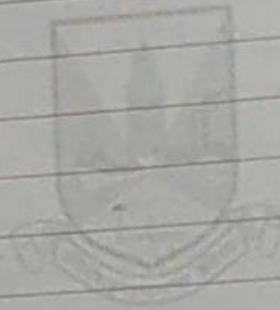
Space for Marks	Question No.
	2. b Certificate base mechanism of Authentication is very best thing in Security.
	→ with help of example we can better understand about certificate base mechanism of Authentication → let's say you are working in a Company there are use certificate base authentication → when you will go for job they ask some document like Marksheet and any other Certificate → they you will give the Marksheet → CA will check it is valid or invalid → CA will go to your organization and getting information about → If it will be prove then CA send the certificate to organization after that you can join the organization.

Space
for
MarksQuestion
No.

→ If any fraud will be ~~compound~~ ~~see~~ certificate related to four ~~provide~~ that time they will not provide the job in this originator

Q.2B-1

→ In simple we can say that they check the certificate is original or fake. If any thing will be fake that time they give loan that job.





Space for Marks	Question No.
	2.C Explain the following terms, ① Hijacking, ② Phishing
	Hijacking:- It is similar to the real world just like plane Hijacking
	→ For Example:- A plane is hijack by the Hijacker. Hijacker want to fulfill some demand from organization or government. If government will not fulfill the demand that time they start killing and beating the passenger.
	→ Hijacking: it is a process in which a virus or malware which insert into the application when it get to change of hijacking they start to hijack the software and starting to demanding software updates. They demand some money for leaving the application.
	- It is a hacker which can be hack the application.

Q.2E - 1

Space
for
MarksQuestion
No.

→ Phishing:- It is also a technique in which they insert in to the software and start to damage the software.

→ It work slowly slowly in the software.

→ ~~It~~ At the last they convert all the part of that software.

→ It is similar to the hijacking but it is very slow process of attacking.

→ We can not find phishing early in our software.

Space
for
MarksQuestion
No.

2.f. Authentication:- Authentication is a process of checking the details of a person which is entering in the system.

→ In Authentication they will check which person is want to enter software storage. It will authenticate or not. means the data is match to verify.

→ If user will pass the Authentication process that he can access the selective data in the system.

→ Process of Smart card implementation

→ In this time there are different types of smart card is present in the market

→ In every smart card have a unique code or pass word

→ During the use of the smart card we have the idea about the code and password of the smart card.

Space
for
MarksQuestion
No.

Q. 2F - 2

→ when you will insert the smart card into the smart card machine that time it ask enter your PIN or password. If you will enter correct PIN or password it will give permission to access all the data about your smart card. If it will with wrong thing we say an invalid input. reenter your PIN.



Space for Marks	Question No.
	3a Explain the following terms:
	① NAT,
	② PAT
	→ NAT:- It is a security related part of that computer which is use protect the. In NAT the data travels in a network and there are different method to protect the data through NAT. Method it is a easy and smart way to protect the data. during networking traveling
	→ PAT:- using PAT we can protect the all sensitive data according our need. but we can not save the data 100%. with any method. So we can not say any of them are fully protected.
	→ some are very protect some are less



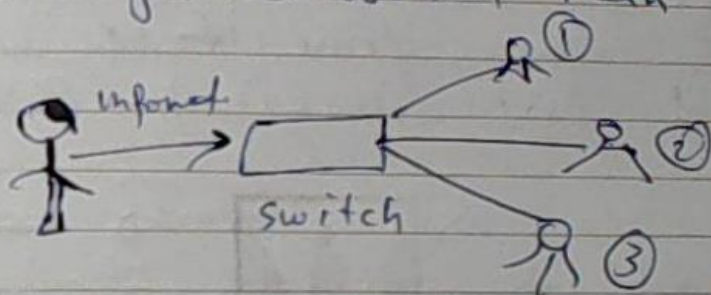
Space for Marks	Question No.
	→ In this to time data protection is very import becuss every thing is going online
	→ Example bussines, education travelling, etc.

Q. 2F - 1

Space
for
MarksQuestion
No.3^e

switch:- in switch, the data is travel through a Network. It is very safe to protect the data:- we can understand switch with example.

→ Let's If am sending an information through switch network



In the example of upper diagram when we send that the data switch send the data only those person where we want to send the information like I send the information to person 3. Switch send the only ③ person. not ①, ② person.

→ It is a prevention of data

→ Hub:- Hub is just opposite of the switch

→ If we want to send the →

Space
for
MarksQuestion
No.

Information to my one

→ information will go all person
• not only a selected person

→ It is a dumb way of sent
informat

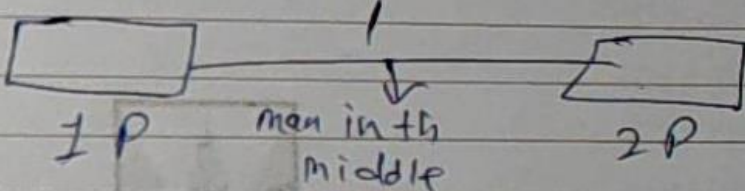
→ but some time it is import
and fast working process

→ If you are a teacher ~~are~~ you
want to tell the student
some important thing that
time, we can use Hubs
method.

Q.3E-1

→ It will take less time
for spreading the information,



Space for Marks	Question No.
	3. f
	There are different form of wireless attacks ?
	Some are here
	→ ① Man in the Middle:- In this type of attack, attacker leave a virus bet ween two data Network  In this diagram the attacker come in to between 2 person and is turn the Network connection.
	✓ ② Bluetooth:- It is also a wireless connection in which attacker attack the connection bluetooth connect if it will connected the person then start send viruses and Malwares
	③ Wifi:- it is similar to the bluetooth but it connect to the internet



Q.3F - 2

Space for Marks	Question No.

→ DDOS :- in this wireless attack attacker attack multiple time at the same server, due to attacking multiple time server can down. that time attacker get the change to enter the system or by using wrong password and starting to damaging the system.



Space for Marks	Question No.	
	4.d.	Network Protocol Attack: It is a one type of attack in which we want to prevent the Network by the attacker. There are different view of point of Network Protocol Attack • Resource Allocation:- we don't know where is the data and informal of that organization. If we get the location of the resource then it will easy to stolen the data. ✓ • Big barrier:- If we want to prevent the source of data we use big barrier. • Virtual protector:- we can use virtual protect that can only seen by by valid person of the organization. • Systematic rule of Network:-



Space for Marks	Question No.
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Q. 4D-1

follow every protection layer.
which can be help to the
source protection. they can
not insert into the main
source of data point



Space for Marks	Question No.
	4. e
	Classic security model is a model that is use in computer networking as security purpose.
	- the classic security model is use to prevent the data from the outsider like Hacker, virus etc
	→ There are different type of classic security Model.
	→ Loop of security:- In this model we can store the data in the form of loop. If any want to access the data they can not easily access due to loop security
	→ Authentication:- In Authentication only those person can access the  data which are authenticated other can not access.
	→ Authorization:- If any person want to access the information about an store first take authority by the organizer of authentication.

Space
for
MarksQuestion
No.

Q. 4E - 1

Randon barrier: it is a gate of any security point if any one want to insert any security point first insert into Randon barrier. possession of that Randon thing.



Space for Marks	Question No.
	4. f Access Control List:- It is a part of computer security in which we have access to point any data by the using of these aspect → In Access Control list we control that Accessibility of the data → <u>Permissibility</u>:- If any one want to access any data of security level first take permission from any valid perm. → <u>Authentication</u>:- to with the use of that we can access the data → <u>Authorization</u>:- If any one is authorized by the organization that time it can access the data. Q. 4F-1 <u>Special control detection</u>:- In this if any one have special control system that time they can access the data



Space
for
Marks

Question
No.

5.6 Some basic factor have to considered in physical security while choosing a site location.

→ There are some factor:-

→ Accessibility:- If the organization is present at the accessible point it can be accessible easy but it will not much accessible.

→ visibility:- It can be at visible place that time no one can stolen any thing

→ lighting:- Proper lighting will provide by the organization

→ Barrier:- Like a House gate it has a barrier

Q.5B-1



→ Physical barrier:- Just like a body guard same as guard will present at the gate

→ High wall system:- It can be covered by the high wall



Space for Marks	Question No.	
	5d.	different phase of security Development life cycle.
	①	Access Control:- first we need to control on the access of all type data.
	②	Security requirement:- In security Requirement we need to collect the data according security Related.
	③	Security Design:- Decide to check How can Model the security
	④	✓ Securing code:- which type of code will be use in the software
Q.5D-1	⑤	Security Test:- in the security test we check all the thing in default of security level.
	⑥	Deployment security:- when we will deploy the software checking the Deployment process.
	⑦	Security maintenance:- all work will done after that maintain the security



Space for Marks	Question No.
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5.e cloud computing security services

→ Cloud computing security services is use to store and easy, and other sensitive work.

→ There are 3 type of cloud computing security

① Infrastructure to service: in this we can use the ~~state~~ cloud computing security service according to the user base point.
✓
Examp (AWS) Amazon web services

② Platform to service: it is a platform like model in this we can use any software in our system.
it can manage by cloud computing

③ Software to service: in cloud computing security services are provided in the form of software we can easily use according to the user needs.

Example Amazon, 365, more soft, tea etc



Space for Marks	Question No.
	→ In this time cloud computing is very very import to understand → Every company need the basic knowles ge about cloud computing → with the help of cloud computing
	Q.5E-1 → we can access the data from any where. and work → cloud computing provide remote work. assestment.

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Space for Marks	Question No.
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Roll No.: 1068899



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BCNO 259617344

Re-Evaluator Sheet

Exam Date

Program Code & Name

Subject Code & Name

Year / Semester

Question wise Marks given by Re-Evaluator

Q. No.	Marks	
1		8
2		9
3		10
4		11
5		12
6		13
7		14
		Total

Name & Signature of Re-Evaluator with Date

Sr. No.

50508512

Program : Science
Seat No. : 1068899
Subject : 88702

Q.No	Marks											
	Main	A	B	C	D	E	F	G	H	I	J	Tot
1		2	NA	2	NA	NA	2					
2		NA	1	NA	NA	1	3					
3		NA	NA	NA	NA	1	2					
4		NA	NA	NA	1	1	1					
5		NA	1	NA	1	1	NA					
Grand Total												